Listing 7.9 `agent_manager.py:BaseAgent`

```
class BaseAgent:
    def __init__(self, chat_history=None):
        self._chat_history = chat_history or []
        self.last_message = ""
        self._actions = []
        self._profile = None

    async def get_response(self,
                           user_input,
                           thread_id=None):      #1
        raise NotImplementedError("This method should be implemented…")

    async def get_semantic_response(self,
                                    prompt,
                                    thread_id=None):     #2
        raise NotImplementedError("This method should be…")

    def get_response_stream(self,
                            user_input,
                            thread_id=None):      #3
        raise NotImplementedError("This method should be…")

    def append_chat_history(self,
                            thread_id,
                            user_input,
                            response):      #4
        self._chat_history.append(
            {"role": "user",
             "content": user_input,
             "thread_id": thread_id}
        )
        self._chat_history.append(
            {"role": "bot",
             "content": response,
             "thread_id": thread_id}
        )

    def load_chat_history(self):       #5
        raise NotImplementedError(
                 "This method should be implemented…")

    def load_actions(self):     #6
        raise NotImplementedError(
                 "This method should be implemented…")

#... not shown – property setters/getters
```

#1 Calls the LLM and returns a response
#2 Executes a semantic function
#3 Calls the LLM and returns a response
#4 Appends a message to the agent's internal chat history
#5 Loads the chat history and allows the agent to reload various histories
#6 Loads the actions that the agent has available to use

Open the `nexus_agents/oai_agent.py` file in VS Code. Listing 7.10 shows an agent engine implementation of the `get_response` function that